MILITARY SPECIFICATION

CHARGE, PROPELLING, 155MM, M3, LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers one type of propelling charge for 155MM howitzer designated as the M3 propelling charge.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL–P–223 —Powder, Black
MIL–A–2550 —Ammunition & Special Weapons, General Specification for
MIL–I–45208 —Inspection Requirements, General Specification for
MIL–I–45607 —Inspection Equipment, Supply and Maintenance of

STANDARDS

MILITARY

MIL–STD–105 —Sampling Procedures and Tables for Inspection by Attributes.
MIL–STD–109 —Inspection Terms and Definitions.

DRAWINGS

ORDNANCE CORPS

76–5–85 —Container, Metal, Propelling Charge Assembly
7548187 —Packing and Marking of Metal Containers, for Propelling Charges, Assembly and Details
8864405 —Charge, Propelling, M3 for 155MM Howitzer, M1 (Green Bag)

PUBLICATIONS

ORDNANCE CORPS

ORD–M608–11 —Procedures and Tables for Continuous Sampling by Attributes.
IEL–8864405 —Index of Inspection Equipment Lists

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with applicable drawings and specifications.

3.2 Propelling charge. The propelling charge shall comply with all requirements specified on Drawing (dwg.) 8864405 and with all requirements specified in applicable specifications.

3.3 Moisture content of black powder. The moisture content of the black powder at the loading station, at the time of loading shall not exceed the amount specified in MIL–P–223, when tested as specified in 4.3.1.

3.4 Propelling charge increment weight. The charge increment weights shall comply with the weights specified in the loading authorization within the following tolerances:

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Increment number	*Tolerance plus or minus (ounces)*
1	0.05
2	0.02
3	0.02
4	0.05
5	0.05

3.5 Chamber gage. The completed propelling charge shall freely enter the prescribed chamber gage.

3.6 Workmanship. All components shall be free of dirt, grease and other foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions. The supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL–STD–109 in order to define terms used herein. The provisions of Specification MIL–A–2550 shall apply.

4.1.1 *Contractor quality assurance system.* The contractor shall provide and maintain an effective quality assurance system in compliance with the requirements of Specification MIL–I–45208.

4.1.2 *Submission of product.* At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

(a) A statement that the lot complies with all quality assurance provisions of the approved current written description of the system.

(b) Number of units of product inspected.

(c) Results obtained for all inspection performed.

(d) Drawing, specification number and date, together with an identification and date of changes.

(e) Certificates of analysis on all material procured directly by the contractor when such material is controlled by Government specifications referenced in any of the contractual documents.

(f) Number of items in the lot.

(g) Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.3 *Government verification.* Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be in accordance with Specification MIL–I–45208 and will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 Inspection provisions.

4.2.1 *Lot formation.* A lot shall consist of propelling charges produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification, and

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same specification revision. Drawing, specification, and process changes not affecting safety, performance or fit, as determined by the Government shall not necessitate changing the lot number. Each lot shall contain:

(a) Propellant from not more than one lot.

(b) Black powder of one interfix lot number from one manufacturer.

4.2.2 *Examination.* Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL–STD–105 except that inspection for critical defects, when listed, shall be 100 percent. Continuous sampling plans in accordance with Handbook ORD–M608–11 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

4.2.2.1 Increment number 1 assembly (see dwg. 8864407 covering a detail of dwg. 8864405).

Categories	*Defects*	*Method of inspection*	*Code No. (see 6.3)*
Critical:			
1.	Igniter charge assembly missing	Visual	01001
Major:	AQL 0.25 percent		
101.	Assembly damaged to the extent that black powder or propellant can escape	Visual	01002
Minor:	AQL 1.50 percent		
201.	Any seam or opening incompletely stitched	Visual	01003
202.	Thread broken	Visual	01004
203.	Straps missing or improperly located	Visual	01005
204.	Evidence of poor workmanship (see 3.6)	Visual	01006

4.2.2.2 Increment loading assembly 2, 3, 4 and 5 (see dwg. 8864406 covering a detail of dwg. 8864405).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical: None defined			
Major:	AQL 0.25 percent		
101.	Assembly damaged to the extent that propellant can escape	Visual	02001
Minor:	AQL 1.50 percent		
201.	Any seam or opening incompletely stitched	Visual	02002
202.	Thread broken	Visual	02003
203.	Evidence of poor workmanship (see 3.6)	Visual	02004

4.2.2.3 Propelling charge (see dwg. 8864405).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Chamber gage failure	Gage	03001
102.	Total length maximum	Gage	03002
103.	Assembly damaged to the extent that black powder or propellant can escape	Visual	03003
104.	Color improper	Visual	03004
Minor:	AQL 2.50 percent		
201.	Increments insecurely tied at junctures	Visual-Manual	03005
202.	Ties loose, broken or insecurely tied	Visual-Manual	03006
203.	Marking misleading or unidentifiable	Visual	03007
204.	Evidence of poor workmanship (see 3.6)	Visual	03008
205.	Igniter protector cap missing or improperly attached	Visual	03009

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4.2.2.4 Container prior to sealing (see dwg. 76–5–85 and 7548187).

Categories	Defects	Method of inspection	Code No.
Critical: None defined			
Major: None defined			
Minor:	AQL 1.50 percent		
201.	Any packing component missing	Visual	04001
202.	Cover cannot be removed by hand	Manual	04002
203.	Charge cannot be removed by hand	Visual	04003

4.2.2.5 Container sealed (see dwg. 76–5–85 and 7548187).

Categories	Defects	Method of inspection	Code No.
Critical: None defined			
Major: None defined			
Minor:	AQL 1.00 percent		
201.	Cover incompletely engaged	Manual	05001
202.	Marking misleading or unidentifiable	Visual	05002

4.2.3 *Testing.*

4.2.3.1 *Moisture content of black powder.* (see 3.3) Major defect. The contractor shall provide adequate controls to insure compliance with the requirements and shall test for verification, at least one sample of black powder from each eight hours production of igniters. If the sample fails to comply with the applicable requirements, the eight hours production represented by the sample shall be rejected. The test shall be performed as specified in 4.3.1, Code No. 06001.

4.2.3.2 *Propelling charge increment weight (see 3.4) and weight of black powder for igniter (see dwg. 8864407). Major defect.* The charges shall be weighed and checked-weighed 100 percent. The check-weighing shall be accomplished independently of the original weighing and, if performed manually, shall be performed by another operator. Any charge which fails to comply with the applicable requirement shall be classed defective and removed from the lot. Equipment shall be in accordance with 4.2.4.

	Code No.
(a) Igniter	07001
(b) Propellant charge	
Increment one	08001
Increment two	08002
Increment three	08003
Increment four	08004
Increment five	08005

4.2.4 *Inspection Equipment.* Index of Inspection Equipment Lists Number IEL 886-4405 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.2.

4.2.4.1 Supply and maintenance of the equipment listed on the IEL shall be in accordance with Specification MIL–I–45607.

4.3 Test methods and procedures.

4.3.1 *Moisture content of black powder.* The moisture content of the black powder shall be determined in accordance with MIL–P–223.

5. PREPARATION FOR DELIVERY

5.1 Packing.

5.1.1 *Level A.* The propelling charges shall be packed in containers complying with dwg. 76–5–85.

5.2 Marking. Marking shall be in accordance with dwg. 7548187.

5.3 Data cards. Data cards shall be prepared for each lot in accordance with the information specified in Specification MIL–A–2550.

6. NOTES

6.1 Ordering data. Procurement documents should specify the title, number, and date of this specification.

6.2 Inspection Equipment. The contractor shall design inspection equipment as required

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by the referenced Equipment List in accordance with the instructions of paragraphs 6.2.1, 6.2.2, 6.2.3, 6.2.4 and 6.2.5.

6.2.1 Inspection Equipment Lists (IEL) indicate the availability of inspection equipment design by showing in the "number" column of the list of inspection equipment—OO Form 1242–3–the numbers of drawings of existing equipment designs, or codes indicating "commercial item" or "contractor design." Design action required of the contractor with respect to the different types of drawings that may be listed is described in paragraphs 6.2.2 and 6.2.3. Action required of the contractor with respect to "commercial item" inspection equipment, is described in paragraph 6.2.4. The contractor will be required to prepare drawings for all equipment coded as "contractor design" in the number column. The code will further indicate whether detailed or specification control drawings are required. These contractor designs must be approved by the Government prior to the fabrication or procuring of the equipment. Designs shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA–ND.

6.2.2 *Detailed drawings.* Detailed drawings are drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. Contractors need provide no design when a detailed drawing is listed for an item of inspection equipment and the contractor elects to use the Government design. The contractor may, however, with the approval of the Government, design alternate inspection equipment to facilitate his operations. Such contractor-prepared designs must be approved by the Government prior to the fabrication or procuring of the equipment. Designs shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA–ND.

6.2.3 *Specification control drawings.* Specification control drawings depict the device in outline, descriptive, diagrammatic, or pictorial form only and specify the required performances or other characteristics. Contractors must prepare drawings of their designs in support of specification control drawings. The IEL code will indicate whether the drawings must be detailed. These contractor prepared designs must be approved by the Government prior to the fabrication or procuring of the equipment. Commercial equipment meeting the requirements of specification control drawings may be approved if described in sufficient detail to permit identification and evaluation by the Government. Designs shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA–ND.

6.2.4 *Commercial items.* Commercial item inspection equipment is equipment that has a universal application for a specific function. It is comprised of items commonly used by industry and government. Contractors are not required to furnish drawings of commercial item inspection equipment, but a list of such equipment must be approved by the Government. Lists shall be submitted for approval to the inspection element of the agency administering the contract.

6.2.5 *Use of standard measuring equipment.* If the contractor desires to use standard measuring equipment in lieu of Government or contractor designed gages, diagrammatic sketches or other descriptions of the use of such equipment shall be submitted to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA–ND, for approval.

6.3 Inspection code numbers. The five-digit code numbers assigned to the inspection herein are to facilitate future data collection and analysis by the Government.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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